Riemann Roch Generalized

Contents

Consider the Gauss-Bonnet theorem (this):

$$\int_M K dA = \int_{\partial M} k_g ds = 2\pi\chi(M)$$

where M is a 2-dimensional compact Riemann manifold, K the Gaussian curvature, k_g the geodesic curvature, and $\chi(M)$ the Euler characteristic. This can be thought of as a special case of The Riemann-Roch Theorem!

There is also a more direct topological generalization of the Gauss-Bonnet theorem known as the Chern theorem, or Chern–Gauss–Bonnet theorem, to $2n$-dimensional Riemann manifolds without boundary:

$$\chi(M) = \int_M e(\Omega)$$

where $\chi(M)$ is the Euler characteristic, Ω is the curvature form (or Levi-Civita connection), and $e(\Omega)$ is the Euler class:

$$e(\Omega) = \frac{1}{(2\pi)^n}\,\mathrm{Pf}(\Omega)$$

where Pf if the *Pfaffian*; namely as Ω can always be represented as a skew-symmetric matrix (when the dimension of M is even, hence the $2n$) it can always be written as the square of a polynomial in matrix entries, namely if A is the skew-symmetric matrix:

$$\mathrm{Pf}(A)^2 = \det(A)$$

for example:

$$A = \begin{pmatrix} 0 & a \\ -a & b \end{pmatrix} \qquad \mathrm{Pf}(A) = a$$

$$B = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix} \qquad \mathrm{Pf}(B) = af - be + dc$$

Atiyah–Singer index theorem seems to generalize what I'm working on further.

Rapid fire concepts we know:

- We shall take varieties and schemes for granted
- We shall be working with algebraic curves, usually algebraic plane curves; hence curves given by $f(x, y)$ for coordinate functions x, y.
- Such curves are projective ([2]). If they are smooth, we naturally ask about their relation to one-dimension complex manifolds; these are called Riemann surfaces (as they are 2 real dimensional)
- Compact Riemann surfaces are classified up to homeomorphism by their geus, and $\chi(X) = 2 - 2g$ where χ is the Euler Characteristic.
- Via normalization, we have the following (This is in fact proven in [5]

Theorem 0.1: Characterizing Riemann Surfaces Using Normalization

Any comapct Riemann surcae $\tilde{C}$ can be obtain through the normalization of a certain plane algebraic curve C wit hat most ordinary double points, i.e. There exists a holomorphic mapping $\sigma : \tilde{C} \to \mathbb{P}^2(\mathbb{C})$ such that $\sigma(\tilde{C})$ is an algebraic curve possessing at most ordinary double points

- The geometry polynomial function (by FTC) makes them surjective and n-to-1 except at the ramification points.
- We shall take holomorphic and meromorphic differential forms for granted
- Key result about holomorphic differential forms:

Theorem 0.2: Stokes Theorem For Holomoprhic Differentials

Let C be a Riemann surface, $\Omega \subseteq C$ an open set, $\overline{\Omega}$ compact and $\partial\Omega = \gamma$ a piecewise smooth curve, and ' a holomorphic differential defined on an open set containing $\overline{\Omega}$. Then:

$$\int_{\partial\Omega} \omega = 0$$

Proof **:**
use local coordinates and apply Cauchy's theorem to get

$$\int_{\partial\Omega_i} \omega = 0$$

on the components, which then gluing gives the desired result

- Residue is winding number

1 Poincaré-Hopf Formula

Let C be a compact Riemann surface. Then we know that for any nonconstant $f \in K(C)$ (the set of meromorphic functions on C), we get:

$$\sum_{p \in C} \nu_p(f) = 0 \tag{1}$$

that is, the number of zeros plus the number of poles (counting multiplicity) is zero. Phillip actually has this really cool generlization of the proof that i got in [3], he prove sthe following:

Theorem 1.1: Residue Theorem For Forms

Let C be a compact Riemann surface and $\omega \in K^1(C)$ a differential form (the K^1 is his notation for diff. form). Then:

$$\sum_{p \in C} \mathrm{Res}_p(\omega) = 0$$

Proof **:**
exercise

Then you can deduce the usual equation (1) by setting $\omega = \frac{df}{f}$. Now, we can ask the follow: for a trivial $\omega \in K^1(C)$, what is:

$$\sum_{p \in C} \nu_p(\omega)$$

A general differential form on a compact Riemann surface has more flexibility than functions, and the order at p is certainly well-defined the same way the residue of a differential form is well defined, so this is a meaningful question. As one-dimensional differential forms are co-vector fields, this is equivalent to a problem on vector-fields, and we know that one interpretation of the Euler-characteristic on surfaces is to count the zero-points (which are sink and source points) and the saddle points (notice how closely this naturally relates to curvature; this is why it comes up naturally in the Gauss-Bonnet Theorem) and so we get:

$$\sum_{p \in C} \nu_p(\omega) = -\chi(C)$$

- There is an interesting word on the schematic representation of curves, [5, p.28]
- Note that meromorphic functions on complex manifold need not be able to map to $\mathbb{C}\mathrm{P}^1$, consider z/w from $\mathbb{C}^2$.
- The end of chapter I, section 10, has a very very simple proof on when one can embedd a Riemann surface into $\mathbb{C}\mathrm{P}^n$!

2 About Normalization

- He starts with a general plane curve, defines singular points to be when the three partial are zero (hence no linear term when translated to $[0:0:1]$, or in his notation $[1:0:0]$). We can then consider any curve $f(x,y) = F(1,x,y)$ when considering the canonical embedding $(x,y) \mapsto [1:x:y]$. Then it must be double/triple/tuple if the lowest degree is ≥ 2, ≥ 3, $\geq k$. It is or ordinary if all the tangent line are distinct.
- We can now consider irreducible components. If it is over a UFD, then the GCD is well-defined and so we get that f, g have no common factors if $(f)+(g) = (1)$ so that if $(f)+(g) = (h)$ then $f = Fh$ and $g = Gh$ so that $fG = gF$. We can detect this using the resultant, and hence we can detect singular points over non-characteristic 0 fields using the discriminant. We can then proceed to show that if C is a curve, S the singular points, and $C^* = C \setminus S$ the compliment of the singular points, then C^* is connected, not necessarily compact, S is finite.
- Next comes normalization which he defines as follows:

Definition 2.1: Normalization

Suppose C is a irreducible plane algebraic curve, S the set of its singular points. Then there exits a compact Riemann surface $\tilde{C}$ and a holomorphic mapping

$$\sigma : \tilde{C} \to \mathbb{C}\mathrm{P}^2$$

such that

1. $\sigma(\tilde{C}) = C$
2. $\sigma^{-1}(S)$ is a finite set
3. $\sigma : \tilde{C} \setminus \sigma^{-1}(S) \to C \setminus S$ is injective

Then $(\tilde{C}, \sigma)$ is the normalization of C, and we write just $\tilde{C}$ if it is unambiguous in context.

- We get the following result equivalent to birational maps:

Lemma 2.2: Dense Injective, then Biholomorphic

Suppose $\tilde{C}, \tilde{C}'$ are Riemann surfaces and

$$h : \tilde{C} \to \tilde{C}'$$

is a surjective holomorphic mapping which is injective on a dense open subset of $\tilde{C}$. Then h is a biholomorphic mapping.

Proof **:**
Prove it is biholomorphic in every neighbourhood.

- Then

Theorem 2.3: Uniqueness of Normalization

The normalization of an algebraic curve C is unique up to isomorphism (it commies in the way you expect)

Proof **:**
use above lemma.

The goal now is to prove a normalization always exists.

- Next Griffith's defines $\mathbb{C}\{x\}$ which is not quite $\mathbb{C}[[x]]$ because it is the set of all *convergent* power series of the form $\sum_k^\infty a_k x^k$. Similarly for $\mathbb{C}\{x, y\}$ which is convergent $\sum_{i,j} a_{ij} x^i y^j$. Then a Weierstrass polynomial is a polynomial of the form. He denotes the later by $\mathcal{O}$.

$$w = y^d + a_1(x) y^{d-1} + \cdots + a_d(x)$$

where

$$a_j(x) \in \mathbb{C}\{x\} \qquad a_j(0) = 0 \qquad j = 1, ..., d$$

Then he goes on to prove the *Weierstrass Propagation theorem* which say that under the appropriate generic conditions , any $f \in \mathcal{O}$ can be written as $f = u \cdot w$ where u is a unit (so $1/u \in \mathcal{O}$ and w is a Weierstrass polynomial.

- Proving this, we get the following great corollary:

Corollary 2.4: Ring Structure of Analytic Func

$\mathcal{O}$ is a UFD, hence any $f \in \mathcal{O}$ can be expressed up to unit u as:

$$f = uf_1 f_2 \cdots f_n$$

where each f_i is irreducible in $\mathcal{O}$.

Proof **:**
Note that proving $\mathbb{C}\{x\}$ is a UFD is easy because any $f(x) = x^m h(x)$ with $h(x)$ a unit with nonzero constant. For the rest checkout [5].

- An important realization is that being irreducible in $\mathbb{C}\{x\}[y]$ and $\mathbb{C}\{x, y\}$ is the same thing.
- Next, let's take an irreducible curve, center the point of interest at $(0, 0)$, represent it as

$$y^n + a_1(x)y^{n-1} + \cdots + a_d(x)$$

Then locally at 0, we may have multiple local irreducible components. Let's take the curve over $\mathbb{C}^2$. Then around 0, if there are multiple components, it will look like multiple disks that are glued together at a point.

- Using this, we can locally normalize, and then use that to globally normalize
- Next Griffith looks at intersection numbers and Bézout Theorem.

Ramification Divisors and the Riemann-Hurwitz Formula

Let $\nu_f(p)$ be the order of a point of a holomorphic mapping $f : C \to C'$ between Riemann surfaces. As such a map is always surjective if it is non-constant, we have $\nu_f(p) \geq 1$. It ramifies if $\nu_f(p) > 1$, and as ramified points are isolated we know that $\{p \in C \mid \nu_f(p) > 1\}$ is finite. Thus we have the well-defined:

Definition 2.5: Ramification Divisor

The *ramification divisor* of f is :

$$R = \sum_{p \in C} (\nu_f(p) - 1)p \in \text{Div}(C)$$

Recall that if $D = \sum_{p \in C} n_p \cdot p$ is a divisor, $\deg D = \sum_{p \in C} n_p$.

Theorem 2.6: Riemann-Hurwitz Formula

Let C, C' be compact Riemann surfaces where:

$$\text{genus}(C) = g \qquad \text{and} \qquad \text{genus}(C') = g'$$

Let $f : C \to C'$ be a nonconstant holomorphic mapping of degree n. Let R e the ramificatio ndivisor of f. Then:

$$\deg R = 2(g + n - ng' - 1)$$

Proof **:**
First, we shall use the fact that on any compact Riemann surface , there exists a nontrivial meromorphic differential. This could be intuitively seen by the correspondence between compact Riemann surfaces and curves, and on any curve we have a non-trivial meromorphic differential.

Hence, choose $\omega \in K^1(C')$, and let $f^*\omega$ be its pullback. Locally we have:

$$\omega = z^\nu$$

and

$$\omega = g(\omega)d\omega$$

Hence:

$$f^*\omega = \nu g(z^\nu)z^{\nu-1}dz$$

and so the order of each point is:

$$\nu_{f^*\omega}(p) = \nu_f(p)\nu_\omega(f(p)) + \nu_f(p) - 1$$

and so the divisor of $(f^*\omega)$ is :

$$\begin{aligned}
(f^*\omega) &= \sum_p \nu_f(p)\nu_\omega(f(p))p + R \\
&= \sum_q \nu_\omega(q)\left(\sum_{f(p)=q} \nu_f(p)p\right) + R \\
&= \sum_q \nu_\omega(q)f^{-1}q + R \\
&= f^{-1}\left(\sum_q \nu_\omega(q)q\right) + R \qquad &= f^{-1}((\omega)) + R
\end{aligned}$$

Thus:

$$\deg(f^*\omega) = \deg f^{-1}((\omega)) + \deg R$$

Then from the Poincaré-Hopf formula:

$$\deg(f^*\omega) = 2g - 2 \qquad \deg(\omega) = 2g' - 2$$

Thus directly computing we get:

$$\begin{aligned}\deg f^{-1}((\omega)) &= \sum_q \nu_\omega(q)\left(\sum_{f(p)=q}\nu_f(p)\right)\\ &= \deg(\omega)\deg f\\ (2g'-2)n\end{aligned}$$

then substituting we get:

$$2g-2=(2g'-2)n+\deg R$$

re-arranging gives our desired result.

From this we can immediately conclude:

Corollary 2.7: Holomorhpic Map and Genus

Let $f : C \to C'$ be a holomorphic map between genus surfaces of genus g, g'.Then if $n \geq 1$:

$$g \geq g'$$

Proof **:**
Apply the formula.

Corollary 2.8: Compact Riemann Surface As Algebraic Equation

Let C be a compact Riemann surface and let

$$x, y : C \to \mathbb{C}\mathrm{P}^1$$

be holomorphic mapping where $\deg x = d$. Then there exists a polynomial equation f such that $f(x, y) = 0$.

Proof **:**
[5, p.94]

Next, we cover the genus formula.

Theorem 2.9: Genus Formula

Let $C \subseteq \mathbb{C}\mathrm{P}^2$ be an irreducible algebraic curve of degree d with δ singular points all being ordinary double points, and let $\sigma : \tilde{C} \to C$ be its normalization where genus$(\tilde{C}) = g$. Then:

$$g = \frac{(d-1)(d-2)}{2} - \delta$$

Proof **:**
[5, p.95].

3 The Riemann-Roch Theorem

Let C be a compact Riemann surface. Define the divisor to be the formal sum:

$$D = \sum_{i}^{k} n_i p_i \in \mathrm{Div}(C)$$

A divisor is called *effective* if each coefficient is ≥ 0. and we write $D \geq 0$. Any divisor can be expressed as the difference of two effective divisors. We now define the Riemann-Roch space:

$$\mathcal{L}(D) = \{f \in K(C) \mid (f) \geq -D\}$$

That is, $\mathcal{L}(D)$ contains all $f \in K(C)$ whether f has a pole of multiplicity at most n_i at p_i if $n_i > 0$ and a zero of multiplicity at least $|n_i|$ at p_i if $n_i < 0$. We can then define:

$$l(D) = \dim \mathcal{L}(D)$$

We then define a similar notion for divisors:

$$K^1(D) = \left\{\omega \in K^1(C) \;\middle|\; (\omega) \geq D\right\}$$

where now, if $n_i < 0$ then ω can have a p_i as a pole of multiplicity at most $|n_i|$ and if $n_i > 0$ then ω must have a p_i as a zero of multiplicity at least n_i. Again , $K^1(D)$ is a vector space and :

$$\dim K^1(D) = i(D)$$

We shall usually be focusing on effect divisors; $D \geq 0$; in that case we have that $\mathbb{C} \subseteq \mathcal{L}(D)$. Furthermore, to be more align with the usual differential geometry notation, we shall write $\Omega^1(D)$ instead of $K^1(D)$. Then for $D \geq 0$, $\omega \in \Omega^1(D)$ means ω is holomorphic and has a zero of appropriate multiplicity at p_i for appropriate points given by D.

Definition 3.1: Linearly Equivalent

Let $D, E \in \mathrm{Div}(C)$. Then D,E are *linearly equivalent* if there exists a $f \in K(C)$ such that

$$(f) = D - E$$

and we write $D \sim E$.

We can also write $D = (f) + E$ for some (f), and hence we are really saying that $D \sim E$ is they are equal up to principal divisor. Hence, two divisor are linearly equivalent if they belong to the same coset in $\mathrm{Div}(C)/\mathrm{PDiv}(C)$.

Proposition 3.2: Dimension and Linear Equivlaence

Let $D, E \in \mathrm{Div}(C)$, $D \sim E$. Then:

$$\mathcal{L}(D) \cong \mathcal{L}(E) \qquad \Omega^1(D) \cong \Omega^1(E) \qquad \deg D = \deg E$$

Proof **:**
Say $D \sim E$. Then there exists an $f \in K(E)$ such that

$$(f) = D - E$$

Now pick $g \in CL(D)$ so that $(g) + D \geq 0$. Then:

$$(fg) + E = (f) + (g) + E = D - E + (g) + E = (g) + D \geq 0$$

Thus, there is a mapping:

$$\mathcal{L}(D) \to \mathcal{L}(D) \qquad g \mapsto fg$$

This mapping is certainly linear, and it is bijective as the inverse is:

$$\mathcal{L}(E) \to \mathcal{L}(D) \qquad g \mapsto \frac{g}{f}$$

Thus, $\mathbb{C}(D) \cong \mathcal{L}(E)$. Similarly, we certainly get that $\Omega^1(D) \cong \Omega^1(E)$. TO show the degree's are the same, recall that for any $f \in K(C)$:

$$\sum_{p \in C} \nu_p(f) = 0$$

Then:

$$\sum \nu_p(f) = \deg(f) = \deg D - \deg E$$

hence $\deg E = \deg D$, as we sought to show.

Proposition 3.3: Brill-Noether Reciprocity

Let $\omega \in \Omega^1(C)$ and $(\omega) = D + E$. Then:

$$\mathcal{L}(D) \cong \Omega^1(E) \qquad \mathcal{L}(E) \cong \Omega^1(D)$$

Proof **:**
Let $f \in \mathcal{L}(D)$ so that $(f) \geq -D¿$ Then:

$$(f\omega) = (f) + (\omega) \geq D + D + E = E$$

Thus define:

$$\mathcal{L}(D) > \Omega^1(E) \qquad f \mapsto f\omega$$

which is certainly a vector-space homomorphism with inverse:

$$\Omega^1(E) \to \mathcal{L}(D) \qquad \varphi \mapsto \frac{\varphi}{\omega}$$

showing they are isomorphic. By symmetry, we have the similar isomorphism by swapping E and D, completing the proof.

Definition 3.4: Laurent Principal Part

Let $\sum_{i=-n}^{\infty} a_i z^i$ be a Laurent series. Then its singular part

$$\sum_{i=-n}^{-1} a_i z^i$$

is the *Laurent principal part*, or just *principal part*. If $a_{-n} \neq 0$, then N is called the order of the Laurent principal part.

Now suppose C is a Riemann surface, D an effective divisor. Let $z_i, i = 1, 2, ..., k$ be local coordinates near p_i and $z_i(p_i) = 0$. Then if for every p_i, there e is given a Laurent principal part of order n_i:

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \cdots + \frac{a_{i1}}{z_i}$$

we can ask the following question: under what condition does there exists an $f \in K(C)$ such that in a neighbourhood of each p_i, f has η_i as a Laurent principal part? This is called the *Mittag-Leffler problem*, and answering it will allow us to deduce the Riemann-Roch Theorem.

First off, a meromorphic function on a compact Riemann surface C is up to a constant uniquely determined by its Laurent principal part, for if $f_1, f_2 \in K(C)$ that have the same Laurent principal part then $f_1 - f_2$ is holomorphic on C, and hence is constant. Thus, we may identify $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$, $d = \deg D$, as follows. Fix $D = \sum_i^k n_i p_i \geq 0$, and let $_i$ be s coordinate function around p_i where $z_i(p_i) = 0$. Then for $f \in \mathcal{L}(D)$, let η_i be the principal part at p_i. Then we know that:

$$\eta_i = \frac{a_{in}}{z^{n_i}} + \cdots + \frac{a_i 2}{z_i^2} + \frac{a_{i1}}{z_i}$$

where $a_{in_i}, ..., a_{i1} \in \mathbb{C}$. Now let P_i be the n_i-dimensional vector space over $\mathbb{C}$ spanned by $\{1/z_i^{n_i}, ..., 1/z_i^2, 1/z\}$. Then we have the injective vector-space homomorphism:

$$\frac{\mathcal{L}(D)}{\mathbb{C}} \to P_1 \oplus \cdots P_k \qquad [f] \mapsto (\eta_1, ..., \eta_k)$$

Then by identifying each P_i with $\mathbb{C}^{n_i}$, we get that this space is a subspace of $\mathbb{C}^d$. Note that surjectivity is not so obvious or even necessary because we don't know if these maps are compatible (ex. the gluing conditions can impose restrictions).

3.1 Dimension of $\Omega^1(C)$

The goal is to show the following:

Theorem 3.5: Dimension of Differential Forms

Let C be a compact Riemann surface of genus g. Then:

$$\dim \Omega^1(C) = g$$

Note that this is easy in the case where $g = 0$, this is the Riemann sphere and we showed that the only holomorphic differential forms on the Riemann sphere are constant. The proof will proceed in two parts:

Proposition 3.6: Dimension of Differential Form: I

Let C be a compact Riemann surface of genus g. Then:

$$\dim \Omega^1(C) \geq g$$

The key idea is that we can "embed" the space of homogeneous functions of the right degree (minus the singularities) into $\Omega^1(C)$ by essentially taking it's differential with a bit of normalization.

***Proof* :**
First, by normalization, for a compact Riemann surface C, there exits a holomorphic mapping $\pi : C \to \mathbb{P}^2$ such that $C' = \pi(C)$ is an irreducible algebraic curve of degree d in $\mathbb{P}^2$ possessing δ ordinary double points, say $p_1, ..., p_\delta$, and no other singular points. Then if $d = 1$ or $d = 2$, then by the genus formula $g = 0$. Thus we shall assume $d \geq 3$.

Now suppose C is mapped holomorphically onto $\mathbb{P}^2$. We may apply a transformation of coordinates so that C' is not argent to the lint at infinity L_∞ in $\mathbb{P}^2$, that L_∞ doe snot pass through the point s $p_1, ..., p_\delta$, and that C' does not pass through $[0 : 0 : 1]$ (so the origin, $(0, 0)$). Indeed, these conditions can be guaranteed due to the following. We can guarantee that C' and L_∞ for we can always find a linear transformation so that C' intersects a line at d distinct points. Taking this line to be L_∞, suppose C' and L_∞ are tangent. Then at the point of tangency, their intersection number would be 2,and so we get by Bézout that $(C' \cdot L_\infty) \geq d + 1$, a contradiction. We can then do a similar reasoning to guarantee $L - \infty$ does not pass through $p_1, ..., p_\delta$. Finally, for the $[0 : 0 : 1]$ condition, take a point which is on L_∞ but not on C', and apply a transformation so that the point becomes $[0 : 0 : 1]$. Then we got our desired coordinates.

We can now write out our affine equation for C' such that $f(x, y) = 0$. Let $\Gamma = p_1 + \cdots + p_\delta$, and let S^{d-3} represent the set of homogeneous polynomials with complex coefficients of degree $d - 3$ in three variables. Let:

$$S^{d-3}(-\Gamma) = \left\{ G(\zeta^0, \zeta^1, \zeta^2) \in S^{d-3} \;\middle|\; G(p_i) = 0, \ \forall i = 1, 2, ..., \delta \right\}$$

Note that by some counting arguments we get:

$$\dim s^{d-3} = \frac{(d-1)(d-2)}{2}$$

On the other hand, as the collection of polynomials that vanish at p_i is equivalent to the fact that the coefficients of this polynomial satisfy a linear equation, we get:

$$\dim S^{d-3}(-\Gamma) \geq \frac{(d-1)(d-2)}{2} - \delta \overset{!}{=} g$$

where $\overset{!}{=}$ comes form the genus formula, theorem 2.9. Thus, if we show there is an injective homomorphism from $S^{d-3}(-\Gamma)$ into $\Omega^1(C)$, we have that:

$$\dim \Gamma^1(C) \geq \dim S^{d-3}(-\Gamma) = g$$

To that end, for any $G \in S^{d-3}(-\Gamma)$. We need to find a form $\omega_G \in \Omega^1(C)$ for it to map to. What we will do is find a pullback of a form α along π. Let $g(x,y) = G(1,x,y)$, $U_0 = \{[1,x,y] \in \mathbb{P}^2\}$ and $U_1 = \{[u,1,v] \in \mathbb{P}^2\}$. Define the following:

$$\alpha_g = \begin{cases} \left.\frac{g(x,y)dx}{\frac{\partial f(x,y)}{\partial y}}\right|_{C'}, & \text{on } C' \cap U_0, \text{ when } \left.\frac{\partial f(x,y)}{\partial y}\right|_{C'} \neq 0 \\ \left.-\frac{g(x,y)dy}{\frac{\partial f(x,y)}{\partial x}}\right|_{C'}, & \text{on } C' \cap U_0 \text{ when } \left.\frac{\partial f(x,y)}{\partial y}\right|_{C'} = 0 \\ \left.-\frac{g\left(\frac{1}{u},\frac{v}{u}\right)\frac{dv}{u^2}}{\frac{\partial f}{\partial y}\left(\frac{1}{u},\frac{v}{u}\right)}\right|_{C'}, & \text{on } C' \cap U_1 \end{cases}$$

where $\frac{\partial f}{\partial y}(1/u, v/u)$ is the value of $\frac{\partial f}{\partial y}$ at $(1/u, v/u)$. As C' doe not has through the point $[0,0,1]$, α is defined over all of C'. It is in fact uniquely defined. Indeed, inside U_0 the affine equation is $f(x,y) = 0$ and thus on C':

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0$$

and thus:

$$\left.\frac{dx}{\frac{\partial f(x,y)}{\partial y}}\right|_{C'} = -\left.\frac{dy}{\frac{\partial f(x,y)}{\partial x}}\right|_{C'}$$

Thus, we have the linear homomorphism:

$$S^{d-3}(-\Gamma) \to K^1(C) \qquad G \mapsto \omega_G = \pi^*(\alpha_g)$$

Note this map is injective since if $\omega_G = 0$, then C' is contained in the curve $g(x,h) = 0$, but this is impossible since $\deg G = d-3$. We must prove that ω_G is indeed holomorphic, and hence it maps into $\Omega^1(C)$, which shall complete the proof.

To that end, we must that there are no poles. If a point $4\text{p}of4\omega_G$ on $\pi^{-1}(C' \cap U_0)$ can become a pole of ω_G, then:

$$\left.\frac{\partial f(x,y)}{\partial y}\right|_{p'} = \left.\frac{\partial f(x,y)}{\partial x}\right|_{p'} = 0$$

where $p' = \pi(p)$. Then p' is a double point of C', so $p' \in \Gamma$, so

$$g(x,y)|_{p'} = 0$$

But p' is a second-order zero of F, and hence a first order zero of $\partial f/\partial x$, so:

$$\alpha|_{p'} = -\left.\frac{g(x,y)dy}{\frac{\partial f(x,y)}{\partial x}}\right|_{C'} \neq \infty$$

hence ω_G is holomorphic at p.

Now consider ω_G on $\pi^{-1}(C' \cap L_\infty)$. Then by definition of α:

$$\alpha = -\left.\frac{u^{d-3}g\left(\frac{1}{u},\frac{v}{u}\right)du}{u^{d-1}\frac{\partial f}{\partial y}\left(\frac{1}{u},\frac{v}{u}\right)}\right|_{C'}$$

to simplify notation, denote the denominator as $h(u,v)$. Then we must show $h(u,v) \neq 0$ on $C' \cap L_\infty$. To see this, note that the equation of C' on U_1 is

$$k(u,v) = u^d f\left(\frac{1}{u}, \frac{v}{u}\right) = 0$$

And notice:

$$\frac{\partial k(u,v)}{\partial v} = u^d \frac{\partial f}{\partial u}\left(\frac{1}{u}, \frac{v}{u}\right) \cdot \frac{1}{u} = h(u,v)$$

With this setup, suppose $p' \in C' \cap L_\infty$ such that

$$h(u,v)|_{p'} = 0$$

Then since p' is not a double point of C', it must be that :

$$\left.\frac{\partial k(,uv)}{\partial u}\right|_{p'} \neq 0$$

Since $p' \in L_\infty$, the coordinates of p' must be of the form $[0,1,v]$. Then we can write the tangent line to C' at p' as :

$$\left.\frac{\partial k(u,v)}{\partial u}\right| |_{p'}(u-0) = 0$$

But then the tangent line is simply L_∞, which contradicts our choice of coordinates, hence no such point exits, and ω_G is holomorphic, as we sought to show.

Let us now show the other direction of $\dim \Omega^1(C) \leq g$. This direction shall require some differential geometry reasoning.

Let C be a compact Riemann surface of genus g with $\gamma_1, \gamma_2, ..., \gamma_{2g-1}, \gamma_{2g}$ framing a $\mathbb{Z}$-basis of the homology group $H_1(C, \mathbb{Z})$. Then if γ is a closed curve on C,

$$\gamma = \sum_{i=1}^{2g} n_i \gamma_i + \partial \Omega$$

where Ω is a region on C. Let $[\gamma]$ represent the homology class containing γ.

Proposition 3.7: Natural Dual of Homology Group

Let γ be a closed differential one-form on C. Then the mapping:

$$\eta_\lambda : H_1(C, \mathbb{Z}) \to \mathbb{C} \qquad [\gamma] \mapsto \int_\gamma \lambda$$

is well-defined

***Proof* :**

The key is that γ is closed. If 'g' is homologous to γ¡ then $\gamma - \gamma' = \partial\Omega$ where Ω is a region on

C. By stokes theorem

$$\eta_\lambda(\gamma) - \eta_\lambda(\gamma') = \int_{\gamma-\gamma'} \lambda = \int_{\partial\Omega} \lambda = \iint_\Omega d\lambda = 0$$

Proposition 3.8: Exactness

Let λ be a closed differential one-form on C. Then if for all i,

$$\int_{\gamma_i} \lambda = 0$$

then λ is exact.

***Proof* :**
We must find f such that $df = \lambda$. Let $p_0 \in C$ be some fixed point. Then for any point $p \in C$, choose a path γ from p_0 to p and define:

$$f(p) = \int_\gamma \lambda$$

If this is well-defined, this is the function we need. Indeed, if γ' is another path from p_0 to p¡ then $\gamma - \gamma'$ is a closed curve on C, and hence:

$$\gamma - \gamma' = \sum_i^{2g} n_i\gamma_i + \partial\Omega$$

where $ni \in \mathbb{Z}$ and Ω is a region on C. Then as:

$$\int_{\gamma_i} \lambda = 0 \qquad \forall i = 1, ..., 2g$$

and

$$\int_{\partial\Omega} \lambda = \iint_\Omega d\lambda = 0$$

we get:

$$\int_\gamma \lambda = \int_{\gamma'} \lambda$$

Thus, f is a well-defined smooth function on C, and by the FTA we get $df = \lambda$, as we sought to show.

Proposition 3.9: Cauchy Riemann For Differential Forms

Let C be a compact Riemann surface, and $\omega, \varphi \in \Omega^1(C)$. Then if:

$$\omega + \overline{\varphi} = df$$

then

$$\omega = \varphi = 0$$

***Proof* :**
Let $\omega = h(z)dz$ and $\varphi = g(z)dz$. Then as C is one-dimensional:

$$\omega \wedge \varphi = 0$$

and

$$\frac{i}{2}\varphi \wedge \overline{\varphi} = |g(z)|^2 \frac{i}{2} dz \wedge d\overline{z} = |g(z)|^2 du \wedge dv$$

We shall show $\varphi = 0$. For the sake of contradiction suppose $\varphi \neq 0$. Then:

$$\frac{/i}{2} \iint\limits_C \varphi \wedge \overline{\varphi} = \iint\limits_C |g(z)|^2 du \wedge dv > 0$$

But then:

$$\varphi \wedge \overline{\varphi} = \varphi \wedge \omega + \varphi \wedge \overline{\varphi} = \varphi \wedge (\omega + \overline{\varphi} = \varphi \wedge df$$

If we show $\iint_C \varphi \wedge df = 0$, we are done. Indeed, notice that:

$$d(f\varphi) = df \wedge \varphi + f \wedge d\varphi$$

As $\varphi \in \Omega^1(C)$, g is holomorphic, and so:

$$\frac{\partial g}{\partial \overline{z}} = 0$$

Thus:

$$\begin{aligned} d\varphi = d(gdz) = dg \wedge dz & \\ &= \left(\frac{\partial g}{\partial z} dz + \frac{\partial g}{\partial \overline{z}} d\overline{z}\right) \wedge dz = 0 \end{aligned}$$

Hence:

$$\varphi \wedge df = -df \wedge \varphi = -d(f\varphi)$$

Note that $f\varphi$ is a differential one-form, so:

$$\iint\limits_C d(f\varphi) = 0$$

But then:

$$\iint\limits_C \varphi \wedge df = -\iint\limits_C d(f\varphi) = 0$$

By a similar argument $\omega = 0$, completing the proof.

Proposition 3.10: Dimension of Differential Form: II

Let C be a compact Riemann surface of genus g. Then:

$$\dim \Omega^1(C) \leq g$$

Proof **:**
For the sake of contradiction suppose $\dim \Omega^1(C) \geq g+1$ so that there are $g+1$ elements $\omega_1, ..., \omega_{g+1} \in \Omega^1(C)$ that are $\mathbb{C}$-linearly independent. Take the equations:

$$\int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda^j \omega_j + \eta^j \overline{\omega}_j) = 0 \qquad i = 1, 2, ..., 2g$$

where each γ_i is given by choosing a $\mathbb{Z}$-basis for $H_1(C, \mathbb{Z})$. By treating $\lambda^1, ..., \lambda^{g+1}, \eta^1 ..., \eta^{g+1}$, then the above is $2g$ linear equations with $2g+2$ unknowns.s This system must then have a trivial solutions, suppose $\lambda_0^1, ..., \lambda_0^{g+1}, \eta_0^1, ..., \eta_0^{g+1}$ i as solution. Then choosing:

$$\omega = \sum_{j=1}^{g+1} \lambda_0^j \omega_j \in \Omega^1(C) \qquad \varphi = \sum_{j=1}^{g+1} \overline{\eta_0^j} \omega_j \in \Omega^1(C)$$

then for all $i = 1, 2, ..., 2g$:

$$k \int_{\gamma_i} (\omega + \overline{\varphi}) = \int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda_0^j \overline{\omega}_j + \eta_0^j \omega_j) = 0$$

This implies it is exact, hence there exists a C^∞ function f such that

$$\omega + \overline{\varphi} = df$$

But then by proposition 3.9 $\omega = \varphi = 0$. But then as $\lambda_0^1, ..., \lambda_0^{g+1}, \eta_0^1, ..., \eta_0^{g+1}$ is a non-trivial solution so that they are not all0, this contradicts the linear independence of $\omega_1, ..., \omega_{g+1}$, as we sought to show.

Theorem 3.11: Dimension of Differential Forms

Let C be a compact Riemann surface of genus g. Then:

$$\dim \Omega^1(C) = g$$

Proof **:**
combine proposition 3.6 and proposition 3.10.

3.2 Two Important Theorems

Let $H_1(C, \mathbb{Z})$ be the collection of closed forms of a genus g Riemann surface. It will have a basis spanned by $2g$ elements. We have a natural dual definition:

Definition 3.12: 1st Cohomology Group

Let C be a compact Riemann surface. Then the first cohomology group of C is:

$$H^1(C, \mathbb{C}) = \mathrm{Hom}_{\mathbb{C}}(H - 1(C, \mathbb{Z}), \mathbb{C})$$

Then we saw in proposition 3.7 that we have a map:

$$\eta : \{\text{closed differential one-forms on } C\} \to H^1(C, \mathbb{C})$$
$$\lambda \mapsto \eta_\lambda$$

and furthermore this map is injective by proposition 3.8. Let $H_{\mathrm{DR}}(C)$ be the equivalence classes of closed one forms mod exact one forms. Then by proposition 3.8, we have in fact that the above η is a map from $H_{\mathrm{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$. Then, by the "Hodge" condition, we have that:

$$\Omega^1(C) \oplus \overline{\Omega^1(C)} \hookrightarrow H_{\mathrm{DR}}(C)$$
$$\omega \oplus \overline{\varphi} \mapsto \omega + \overline{\varphi}$$

is also injective. We thus get a triple of injective maps:

$$\Omega^1(C) \oplus \overline{\Omega^1(C)} \hookrightarrow H_{\mathrm{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$$

Note that the vector spaces on both ends are $2g$, and hence all these spaces are isomorphic!The first isomorphism is a special case of the *Hodge Theorem*, and the second isomorphism is a special case of *DeRham's Theorem*.

Definition 3.13: Period Vector

Let C be a compact Riemann surface of genus g, $\omega_1, ..., \omega_g$ a basis for $\Omega^1(C)$, and $\gamma - 1, ..., \gamma_{2g}$ a $\mathbb{Z}$-basis for $H_1(C, \mathbb{Z})$.Then the $2g$ g-dimensional vector:

$$\pi_i = \begin{pmatrix} \int_{\gamma_i} \omega_1 \\ \vdots \\ \int_{\gamma_i} \omega_g \end{pmatrix}$$

are called a *system of period vectors of C*. The matrix:

$$\Omega = (\pi_1, ..., \pi_{2g})_{g \times 2g}$$

is called the *period matrix* of C

Proposition 3.14: Period Matrix Nonsingular

The $2g$ period vectors are $\mathbb{R}$-linearly independent.

Proof **:**
For the sake of contradiction, suppose $\pi_1, ..., \pi_{2g}$ are $\mathbb{R}$-linearly dependent so that there exits $a_1, ..., a_{2g} \in \mathbb{R}$ not all zero st

$$a_1\pi_1 + \cdots + a_{2g}\pi_{2g} = 0$$

We make take the complex conjugate of this (note conjugating an integral brings the conjugation to the inside) and get the matrix:

$$\Omega^* \begin{pmatrix} \pi_1 & \cdots & \pi_{2g} \\ \overline{\pi}_1 & \cdots & \overline{\pi}_{2g} \end{pmatrix}_{2g\times 2g}$$

By our assumption, $\text{Rank}(\Omega^*) < 2g$, hence there is a vector in the kernel

$$\int_{\gamma_i} \left(\sum_{j=1}^{g} \lambda^j \omega_j + \sum_{j=1}^{g} \eta^j \overline{\omega}_j \right) = 0 \qquad (i = 1, .., 2g)$$

Let $\omega = \sum_{j=1}^{g} \lambda^j \omega_j$ and $\varphi = \sum_{j=1}^{g} \overline{\eta}^j \overline{\omega}_j$. Then:

$$\int_{\gamma_i} (\omega + \overline{\varphi}) = 0 \qquad (i = 1, ..., 2g)$$

But then we know $\omega = \varphi = 0$, contradicting the linear independence of $\omega_1, ..., \omega_g$, as we sought to show.

3.3 Riemann Inequality

Theorem 3.15: Riemann Inequality

Let C be a compact Riemann surface of genus g, $D \in \text{Div}(C)$, and $\deg D = d$. Then;

$$l(D) \geq d - g + 1$$

see [5, p.113] for full details.

Proof **:**
Like before, let us link Riemann surfaces to curves to get the required context. Let

$$\pi : C \to C' \subseteq \mathbb{CP}^2$$

be the normalization map with δ ordinary double points $p_1, ..., p_\delta$. Griffith's offers the following visualization:

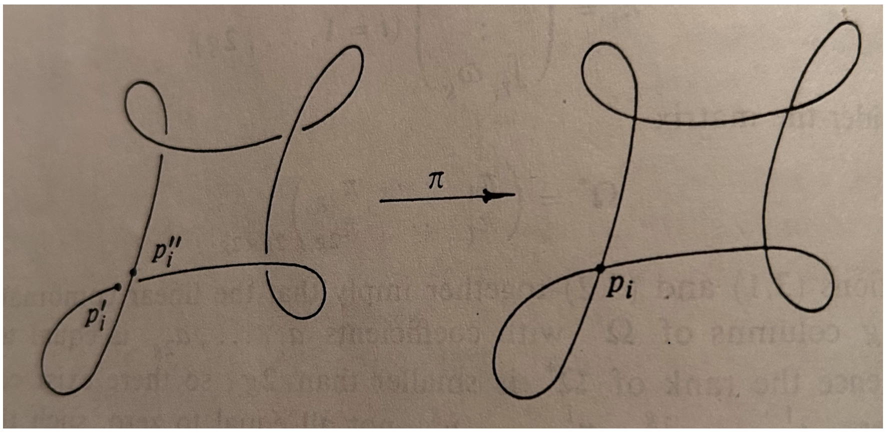

Figure 1: normalizatoin

Let $\pi{-}1(p_i) = p_i' + p_i''$. Then let:

$$\Delta = \sum_{i=1}^{\delta} (p_i' + p_i'') \in \mathrm{Div}(C)$$

Then by a change of coordinates in $\mathbb{C}\mathrm{P}^2$, without loss of generality we may assume neither $\pi(D)$ nor $\pi(\Delta)$ contain the point at infinity (like in the above figure). Now, let C' be given by the equation $F(\xi^0, \xi^1, \xi^2) = 0$, $\deg F = m$. Let $D = D' - D''$ where $D', D'' \geq 0$ and let

$$\deg D' = d' \qquad \deg D'' = d''$$

Let S^n be the vector-space of complex homogeneous polynomials in three variables of degree n. Then:

$$\dim S^n = \frac{1}{2}(n+1)(n+2)$$

Now let's consider a polynomial $G(\xi^0, \xi^1, \xi^2) \in S^n$ satisfying:

1. $F \nmid G$
2. $G \cdot C \geq D' + \Delta$ where $G \cdot C = (\pi^*(G(\xi^0, \xi^1, \xi^2)))$ (that is, the divisor induced by the zeros of $\pi^*(G)$ on C). Note that $G \cdot C \in \mathrm{Div}(C)$.

We shall elaborate why we would want such a polynomial, then show such a polynomial exists. First, for $p \in C$, let $p' = \pi(p) \in C'$. Then a neighbourhood of p gets mapped under π into a curve component of C' near p'; denote it by B_p. Then after a possible coordinate change, we may take p' to be $[1 : 0 : 0]$. Label the equation after this coordinate change G_p. As the multiple points of C' are ordinary double points, the equation of B_p near p' is:

$$y = a_0 + a_1 x^2 + \cdots$$

Putting this into $G_p(1, x, y)$, we get a power series in terms of x. Now, let $n(p)$ denote the degree of the lowest term. We then get:

$$G \cdot C = \sum_{p \in C} n(p) p$$

the right hand side is certainly a finite sum. (I'll finish rest later)

Corollary 3.16: Comparing Degree and Genus

If $d \geq g$, then
$$l(D) \neq 0$$
Hence if $l(D) = 0$, then:
$$d \leq g - 1$$

Proof **:**
immediate

Proposition 3.17: Meromorphic Differentials and Riemann Inequality

$$i(D) \geq g - d - 1$$

Proof **:**
Let F, m, D', D'', d', d'', and Δ be as in theorem 3.15. finish here.

Corollary 3.18: Comparing Degree and Genus – Meromorphic Differential

If $d \leq g - 2$, then $i(D) \neq 0$, hence if $i(D) = 0$ then
$$d \geq g - 1$$

Proof **:**
immediate

Finally this leads us to:

Theorem 3.19: Riemann-Roch Theorem

Let C be a compact Riemann surface of genus g and :
$$D = \sum_{i=1}^{k} n_i p_i \in \mathrm{Div}(C)$$
Then:
$$l(D) = d - g + i(D) + 1$$

Proof **:**
see [5, p.118]

Let us first assume $D \geq 0$ and prove

$$l(D) \leq d - g + i(D) + 1$$

And then take cases and judiciously use the Riemann inequality and its corollary to get the results. Suppose $\eta = (\eta_1, ..., \eta_k)$ where:

$$\eta_i = \frac{a^{in_i}}{z_i^{ni}} + \cdots + \frac{a_{i1}}{z_i}$$

is a Laurent principal part near p_i. The linear space consisting of all these η is isomorphic to $\mathbb{C}^d$. Then thinking of η as a vector in $\mathbb{C}^d$, consider:

$$\begin{aligned}\text{Res} : \mathbb{C}^d \times \Omega^1(C) \to \mathbb{C} \\ (\eta, \omega) \mapsto \sum_{i=1}^{k} \text{Res}_{p_i}(\eta_i\omega)\end{aligned}$$

This is certainly a bilinear mapping. Let us show that:

$$\text{Res}(\eta, \omega) = 0, \ \forall \eta \in \mathbb{C}^d \qquad \text{if and only if} \qquad \omega \in \Omega^1(D)$$

If $\omega \in \Omega^1(D)$, then $\eta_i\omega$ does not have p_i as a pole for all $i = 1, ..., k$, hence:

$$\text{Res}_{p_i}(\eta_i\omega) = 0$$

and so the sum of residues is zero, giving the $\Leftarrow$ direction. For the $\Rightarrow$ direction, assume otherwise so that $\omega \in \Omega^1(C)$ such that $\text{Res}(\eta, \omega) = 0$ for all $\eta \in \mathbb{C}^d$ but $\omega \notin \Omega^1(D)$. Then near p_i, let

$$\omega = b_{i0}z_i^{m_i} + b_{i1}z_1^{m_i+1} + \cdots$$

where $m_i \in \mathbb{Z}$, $b_{i0} \neq 0$. By assumption, $m_i < n_i$. Then there is a Laurent Principal part near p_i

$$\xi_i = (\xi_{i1}, ..., \xi_{ik})$$

such that

$$\xi_{ii} = \frac{1}{z_i^{m_i+1}} \qquad \xi_{ij} = 0 \ (i \neq j)$$

As $m_i + 1 \leq n_i$, $\mathbb{C}^d$, by treating these as $\mathbb{C}^d$ we have $\xi_i \in \mathbb{C}^d$. However:

$$0 = \text{Res}(\xi_i, \omega) = \text{Res}_{p_i}\frac{b_{i0}}{z_i} = b_{i0} \neq 0$$

a contradiction. Thus, $m_i \geq n_i$, and so $\omega \in \Omega^1(D)$.

Next, recall the following fact (or see exercise ref:HERE). Let $B : V \times W \to \mathbb{C}$ be a bilinear map between two finite dimensional vector spaces, in particular $\dim V = n$ and $\dim W = m$.

Then B is called non-degenerated if given $w \in W$, $B(v, w) = 0$ for all $v \in V$ implies $w = 0$. Then given $L \subseteq V$ satisfies $B(L \times W) = 0$, then:

$$\dim L \leq n - m$$

Then by the property we just found earlier about the residue map, we have a non-nondegenerate bilinear map:

$$\mathrm{Res}_1 : \mathbb{C}^d \times \frac{\Omega^1(C)}{\Omega^1(D)} \to \mathbb{C}$$

Now recall we can interpret $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$, and by what we've just shown:

$$\mathrm{Res}_1 \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \times \frac{\Omega^1(C)}{\Omega^1(D)} \right) = 0$$

Then:

$$\dim \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \right) \leq \dim \mathbb{C}^d - \dim \left(\frac{\Omega^1(C)}{\Omega^1(D)} \right)$$

That is:

$$l(D) - 1 \leq d - \dim \Omega^1(C) + i(D)$$

By theorem 3.5, $\dim \Omega^1(C) = g$, and so we get:

$$l(D) \leq d - g + i(D) + 1$$

This was the hardest part of the proof. We now cases cases on $l(D)$ and $i(D)$ to finish:

1. Consider $l(D) \neq 0$, $i(D) = 0$. As $l(D) \neq 0$, there exists an $f \in K(C)$ such that $(f)+D \geq 0$. Let $E = (f) + D$. Then $(f) = E - D$, so $E \sim D$. Then

   $$l(D) = l(E) \qquad i(D) = i(E) \qquad \deg D = \deg E$$

   Thus, we may assume $D \geq 0$. Thus we get our inequality, and by the Riemann inequality:

   $$d - g + i(D) + 1 = d - g + 1 \overset{\text{R.}}{\leq} l(D) \leq d - g + 1$$

   giving equality

2. Consider $l(D) \neq 0$, $i(D) \neq 0$. As $l(D) \neq 0$, we may do the same trick to assume $D \geq 0$. Since $i(D) \neq 0$, there exits a $\omega \in \Omega^1(C)$ such that

   $$(\omega) \geq D$$

   Let $E = (\omega) - D$ so that $E \geq 0$. Then by proposition 3.3 and proposition 3.17 we get:

   $$i(D) = l(E) \leq \deg E - g + 1 + i(E)$$

   By the Poincaré-Hopf formula, $\deg(\omega) = 2g - 2$ so

   $$\deg E = \deg(\omega) - \deg(D) = 2g - 2 - d$$

   As $i(E) = l(D)$, we get:

   $$i(D) \leq 2g - 2 - d - g + 1 + l(D)$$

   re-arranging and combining we get:

   $$l(D) \geq d - g + i(D) + 1$$

   giving us the other wise

3. Consider $l(D) = 0$, $i(D) \neq 0$. Since $i(D) \neq 0$, there exists a $\omega \in \Omega^1(C)$ such that $(\omega) = D + E$ where $E \geq 0$. Then by proposition 3.3

$$l(E) = i(D) \neq 0 \qquad i(E) = l(D) = 0$$

Thus, by case (1) applied to E we get:

$$l(E) = \deg E - g + i(E) + 1$$

Note that

$$\deg E = \deg(\omega) - \deg D = 2g - 2 - d$$

Then substituting and rewriting $l(E)$, $i(E)$ as $i(D)$ and $l(D)$, we get the equality.

4. Consider $l(D) = 0$, $i(D) = 0$. Then by corollary 3.16 and corollary 3.18

$$d = g - 1$$

which is the desired equation.

(then do the exercises for $C = \mathbb{P}^1$ direct prove, and the elliptic curve exercise)

4 Application of Riemann-Roch Theorem

It shall cover genus $0, 1, 2, 3, 4$.

4.1 Genus 0

Proposition 4.1: Genus 0 then Rational

Let C be a compact Riemann surface of genus 0. Then

$$C \cong \mathbb{P}^1$$

***Proof* :**
Let $p \in C$, $D = (p) \in \mathrm{Div}(C)$. Then:

$$0 \leq i(D) \leq i(0) = \dim \Omega^1(C) = 0$$

Thus $i(D) = 0$. By the Riemann-Roch theorem:

$$l(D) = d - g + i(D) + 1 = 1 - 0 + 0 + 1 = 2$$

Thus, $\mathcal{L}(p)$ contains two linearly independent function, one of which can be chosen to be $f_1 = 1$. The other function must be a meromorphic function (as C is a compact Riemann surface) and must have p as a pole of order 1; call this function f_2. Then by definition $f_2^{-1}(\infty) = p$. Hence it follows that this is a holomorphic mapping:

$$f_2 : C \to \mathbb{P}^1$$

where $\deg f_2 = \#\{f_2^{-1}(\infty)\} = 1$ implying that f_2 is in fact biholomorphic (recall lemma 2.2). Thus:

$$C \cong \mathbb{P}^1$$

as we sought to show.

4.2 Genus 1

Proposition 4.2: Genus 1 Weierstrass Equation Representation

Let C be a compact Riemann surface of genus 1. Then C can be represented by a smooth algebraic curve of degree 3 in $\mathbb{P}^3$

***Proof* :**

We shall construct a holomorphic injective map $f : C \hookrightarrow \mathbb{P}^2$ where f shall be a smooth algebraic curve of degree 3. First, for any nonzero $\omega \in \Omega^1(C)$, by Poincaré-Hopf formula we have:

$$\deg(\omega) = 2g - 2 = 2 - 2 = 0$$

Hence, if $\omega(p) = 0$, then it would have to have a pole, but it is holomorphic! Thus, for any $D > 0$, $i(D) = 0$. Thus, letting $D = 2p$, we get:

$$i(D) = 0$$

Thus by the Riemann-Roch theorem we have that:

$$l(2p) = d - g + i(2p) + 1 = 2 - 1 + 0 + 1 = 2$$

Hence $\mathcal{L}(D)$ has non-constant meromorphic functions which only have p as a singular point of order ≤ 2. In fact, it *must* be of order 2, as by the Riemann-Roch theorem:

$$l(p) = 1 - 1 + 1 = 1$$

Hence $\mathcal{L}(p)$ only has constant functions; and so the order of any meromorphic function with single pole p is at least 2. Let $x \in \mathcal{L}(2p)$ be such a function. Then near p we have:

$$x = \frac{c_2}{t^2} + \frac{c1}{t} + h_1(t)$$

for a local coordinate near p where $t(p) = 0$ and $c_2 \neq 0$. Let's start by showing that it must be that $c_1 = 0$. As $\dim \Omega^1(C) = g = 1$, take a $\omega \in \Omega^1(C)$ that is nonzero so that under t

$$\omega = g(t)dt \qquad g(0) \neq 0$$

Then $x\omega$ has no poles other than p, so by the Residue Theorem (the sum of all residues is 0):

$$0 = \mathrm{Res}_p(x\omega) = c_1$$

Thus, with a suitable change of coordinates we get:

$$x = \frac{1}{z^2} + h(z)$$

Note that this x has many similar properties to the Weierstrass function, as we shall see later.

Now, for any $\omega \in \Omega^1(C)$, let
$$y = \frac{dx}{\omega}$$
which is well defined as $\omega(p) \neq 0$ for all $p \in C$. Define:
$$f : C \to \mathbb{P}^2$$
$$q \mapsto [1 : x(q) : y(q)] \qquad (p \mapsto [0 : 0 : 1])$$

Note that our chosen point p was mapped to the point at infinity. Certainly f is holomorphic, we must just show it is injective and that $f(C)$ is a curve of degree 3.

Starting by showing $f(C)$ is a degree 3 curve in $\mathbb{P}^2$, consider $x : C \to \mathbb{P}^1$. Then by construction $x^{-1}(\infty) = 2p$. Thus:
$$\deg x = 2$$

Thus, the ramification can never exceed 1, and hence the ramification index at its ramification points never exceeds 1. In particular, the coefficients of every point for the ramification divisor of x must be 1. By the Riemann-Hurwitz formula;
$$\deg R = 2(\deg x + g - 1) = 2(2 + 1 - 1) = 4$$

As p ramified, there are 3 other Ramification points:
$$R = p_1 + p_2 + P_3 + p$$
each p_i being mutually distinct:

FIG HERE

let $x(p_i) = a_i$. Let us figure out what type of ramification we have. To be succinct with tracking roots and poles, for a meromorphic differential form $\alpha \in K^1(C)$, let $(\alpha)_0$ and $(\alpha)_\infty$ represented the zeros and poles (including multiplicity) respectively. As ω has no zeros or poles then since $y = dx/\omega$ we get:
$$(y)_0 = (dx)_0 \qquad (y)_\infty = (x)_\infty$$

Hence, $(y)_0$ and $(y)_\infty$ can only contain the points of x with ramification index 1, that is p_1, p_2, P_3, p. Thus, each of these "zero" and "pole" divisors are shared between the two.

To calculate, suppose $z_i (i = 1, 2, 3)$ are local coordinates near p_i where $z_i(p_i) = 0$. Since the multiplicity of x at p_i is 2,
$$x - a_i = z_i^2 h_i(z_i)$$
where h_i is a holomorphic function and $h_i(0) \neq 0$. Thus:
$$dx = 2z_i h_i(z_i)dz_i + z_i^2 h_i'(z_i)dz_i$$
$$z_i(2h_i(z_i) + z_i h_i'(z_i))dz_i$$

But then evaluating at $z_i = 0$:
$$(2h_i(z_i) + z_i h_i'(z_i))|_{z_i=0} = 2h_i(0) \neq 0$$

and so p_i is a simple zero of dx. For the pont p, choosing a local coordinate function z near p as we did before we get:

$$x = \frac{1}{z^2} + h(z)$$

Neurally, dx has p as a pole of order 3, and so we found:

$$(y)_0 = p_1 = P_2 + P_3 \qquad (y)_\infty = 3p \qquad (y) = p_1 + p_2 + p_3 - 3p$$

We now know the roots/pole information our desired function must have. Let $g(x) \in K(\mathbb{P}^1)$ be given by:

$$g(x) = (x - a_1)(x - a_2)(x - a_3)$$

Considering g as a function on C (that is x^*g), then as near the points p_i and p in coordinates z_i and z:

$$x - a_i = z_i^2 h_i(z_i) \qquad x = \frac{1}{z^2} h(z)$$

we get:

$$(g)_0 = 2p_1 + 2p_2 + 2p_3 \qquad (g)_\infty = 6p$$

since each p_i takes p_i as a zero of order 2, an near p every $(x - a_i)$ takes p as pole of order 2.

Now, we see that the divisor of g is just the divisor of y^2, and so:

$$\left(\frac{y^2}{g}\right) = 0$$

showing y^2/g is holomorphic on C, and as C is compact $y^2/g = a \in \mathbb{C}$ where $a \neq 0$. To choose a better constant replace y by $y/\sqrt{a}$. Then we get:

$$y^2 = g(x)$$

But now, we just showed that under f that C lies in the image of a curve of degree 3:

$$C' = \left\{[1 : x : y] \in \mathbb{P}^2 \;\middle|\; y^2 - g(x) = 0\right\} \cup \{[0 : 0 : 1]\}$$

Giving us the degree. Let us now check that f is an injective mapping except at ∞.

To that end, we shall need a bit more structure on C. We shall define a natural involution map on C. Let

$$\begin{aligned} j : C &\to C \\ q_1 &\mapsto q_2 \end{aligned}$$

where $q_1, q_2 = x^{-1}(p_r)$ are an arbitrary $p_r \in C'$ (where I chose subscript r for no particular reason, we're running out of letters). Then certainly $j^2 = \mathrm{id}$. It is thus it's own inverse, and so it is easy to verify it is an automorphism. Using this, we shall show some properties of j^* on $K(C)$, $K^1(C)$ and $\Omega^1(C)$. Certainly by definition of j:

$$j^*(x) = x \qquad j^*(d) = dx$$

Let's show $j^*y = -y$. As $y = dx/\omega$:

$$j^*y = \frac{j^*(dx)}{j^*(\omega)} = \frac{dx}{d^*(\omega)}$$

Hence, we show $j^*(\omega) = -\omega$. As $\dim \Omega^1(C) = g = 1$, we get $\Omega^1(C) \cong \mathbb{C}$. Then notice that j can be thought of as a linear transformation on $]C$ so that $j^2 = 1$ implies $(j^*)^2 = 1$, (hence it is diagonalizable) hence the eigenvalues of j^* are either 1 or -1. Suppose j^* has eigenvalue 1 so that $j^*\omega = \omega$. Then there exists a holomorphic differential φ on $\mathbb{P}^1$ such that

$$\omega = \varphi \circ x = x^*\varphi$$

(exercise). But there is no nonzero holomorphic differential on $\mathbb{P}^1$, hence $j^*\omega = -\omega$.

With the properties of j established, consider $q_1, q_2 \in C$ where $f(q_1) = f(q_2)$. We want to show that $q_1 = q_2$. Naturally if either q_1, q_2 is p then by construction $f(q_1) = f(q_2) = \infty$ and so $q_1 = q_2 = p$; thus we assume neither q_1, q_2 is p. By definition:

$$f(q_1) = [1 : x(q_1) : y(q_1)]$$
$$f(q_2) = [2 : x(q_2) : y(q_2)]$$

and as these are equal $x(q_1) = x(q_2)$ and $y(q_1) = y(q_2)$. Now if $q_1 \neq q_2$, as $x(q_1) = x(q_2)$ it must be that $j(q_1) = q_2$. Thus, we have by definition of the pullback:

$$j^*y(q_1) = y(q_2)$$

however as $j^*(y) = -y$

$$j^*y(q_1) = -y(q_1) = -y(q_2)$$

and so

$$y(q_2) = -y(q_2)$$

telling us $y(q_2) = 0$. But $(y)_0 = p_1 + p_2 + p_3$, so $q_2 \in \{p_1, p_2, p_3\}$. Thus:

$$q_1 = j(q_2) = q_2$$

but this contradicts $q_1 \neq q2$, thus it must be that they are equal, showing f is injective.

Let's now show surjectivity. We shall use a connectedness argument. First, $y^2 - g(x) = 0$ is certainly an irreducible polynomial, hence it forms an irreducible subset, hence a connected subset. Next, C is compact, and as f is holomorphic $f(C)$ is closed. As f is holomorphic, $f(C)$ is also open (by the holomorphic open mapping theorem), and so $f(C) = C'$.

Finally, we prove C' is smooth. We can show this directly by computing . By everything shown so far, we know that the projective equation of C' can be written as:

$$F = \xi^0(\xi^2)^2 - (\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0) = 0$$

Thus:

$$\begin{aligned}\frac{\partial F}{\partial \xi^0} = (\xi^2)^2 &+ a_1(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0)\\ &+ a_2(\xi^1 - a_1\xi^0)(\xi^1 - a_3\xi^0)\\ &+ a_3(\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)\end{aligned}$$

$$\begin{aligned}\frac{\partial F}{\partial \xi^1} = &-(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0)\\ &- (\xi^1 - a_1\xi^0)(\xi^1 - a_3\xi^0)\\ &- (\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)\end{aligned}$$

$$\frac{\partial F}{\partial \xi^2} = 2\xi^0\xi^2$$

If $\partial F/\partial \xi^i$ $(i = 1, 2, 3)$ are all zero, by the Euler formula, $F = 0$. Now from

$$2\xi^0\xi^2 = 0$$

we get

$$\xi^0(\xi^2)^2 = 0,$$

and substituting this expression into $F = 0$, we also get

$$(\xi^1 - a_1\xi^0)(\xi^1 - a_2\xi^0)(\xi^1 - a_3\xi^0) = 0.$$

At least one of the three factors on the left side of this expression is therefore equal to 0. Having only one factor equal to 0 however, contradicts

$$\frac{\partial F}{\partial \xi^1} = 0.$$

Thus at least two of the factors are equal to 0. But then $\xi^0 = \xi^1 = 0$ because a_1, a_2, a_3 are mutually distinct. Substituting into

$$\frac{\partial F}{\partial \xi^0} = 0,$$

we get $\xi^2 = 0$, i.e., we get

$$\xi^0 = \xi^1 = \xi^2 = 0,$$

and this is impossible. Thus, C' is smooth. As f is biholomorphic, we have that C is biholomorphic to a smooth degree 3 curve, as we sought to show.

4.3 Genus ≥ 2: Canonical Map

In higher dimensions, there are generally many kinds of mapping from a compact Riemann surface into an n-dimensional projective space. However all compact Riemann surface of genus $g \geq 2$, they exits as special holomorphic mapping from this compact Riemann surface into $\mathbb{P}^{g-1}$ that is called he *canonical map*.

Definition 4.3: Canonical Map

Let C be a compact Riemann surface of genus $g \geq 2$ an $\{\omega_1, ..., \omega_g\}$ a basis for $\Omega^1(C)$. Then:

$$\varphi_K : C \to \mathbb{P}^{g-1}$$
$$p \mapsto [\omega_1(p) : \cdots : \omega_g(p)]$$

Note that this is well-defined. It is certainly not coordinate dependent, and if, for the sake of contradiction, there is a p where $\omega_i(p) = 0$ for all p (which implies p maps to $[0 : \cdots : 0]$ which doesn't exist), then notice that $\Omega^1(p) = \Omega^1(C)$ so

$$i(p) = \dim \Omega^1(C) = g$$

and by the Riemann Roch-theorem

$$l(p) = 1 - g + g + 1 = 2$$

and so there exits nonconstant meromorphic function $f \in \mathcal{L}(p)$ where $f^{-1}(\infty) = p$. Thus

$$\deg f = 1$$

giving us that $f :\to \mathbb{P}^1$ is an isomorphism, but that is not possible as $\mathbb{P}^1$ has genus 0.

Proposition 4.4: Non-degeneracy of Canonical Map

Let φ_K be the canonical map. Then φ_K is non-degenerate.

***Proof* :**
Assume φ_K is degenerate so that there exists $\lambda_\alpha \in \mathbb{C}$ not all zero st for all $p \in C$

$$\sum_{\alpha}^{g} \lambda_\alpha \omega_\alpha(p) = 0$$

Then:

$$\sum_{\alpha}^{g} \lambda_\alpha \omega_\alpha$$

contradicting that $\{\omega_1, ..., \omega_g\}$ is a basis of $\Omega^1(C)$, as we sought to show.

Proposition 4.5: Degree 2 Mapping Condition

Let C be a Riemann surface of genus $g \geq 2$ and let φ_K be the canonical mapping. Then if

$$\varphi_K(p) = \varphi_K(q)$$

for some distinct $p, q \in C$, then there exists a holomorphic mapping of degree 2 from C into $\mathbb{P}^1$

***Proof* :**
If $\varphi_K(p) = \varphi_K(q)$,then $\omega_\alpha(p) = \lambda\omega_\alpha(q)$ for all $\alpha = 1, ..., g$ for some $0 \neq \lambda \in \mathbb{C}$. Let $D = p + q$. Let us find $i(D)$. We shall start by showing that:

$$\Omega^1(D) = \Omega^1(p)$$

Indeed, for any $\omega \in \Omega^1(C)$, if

$$\omega = \sum_{\alpha=1}^{g} \mu_\alpha \omega_\alpha$$

for $\mu_\alpha \in \mathbb{C}$, $\alpha = 1, ..., g$, then:

$$\begin{aligned} \omega(p) = 0 &\Leftrightarrow \sum_{\alpha=1}^{g} \mu_\alpha \omega_\alpha(p) = 0 \\ &\Leftrightarrow \sum_{\alpha=1}^{g} \lambda \mu_\alpha \omega_\alpha(q) = 0 \\ &\Leftrightarrow \omega(q) = 0 \end{aligned}$$

Thus, $\Omega^1(p) = \Omega^1(p+q) = \Omega^1(D)$. Hence, to calculate $i(D)$ it suffices to calculate $i(p)$.

To that end, consider :

$$\lambda_2\omega_1(p) + \lambda_2\omega_2(p) + \cdots + \lambda_g\omega_g(p) = 0$$

for $\lambda_1, ..., \lambda_g \in \mathbb{C}$. Note not all these could be zero or else we would have a point map to $[0 : \cdots : 0]$. Hence, this linear equation *must* have $g-1$ linearly independent solution vectors $(\lambda_1, ..., \lambda_g) \in \mathbb{C}^g$, giving rise to $g-1$ linearly independent elements in $\Omega^1(p)$. Thus:

$$i(D) = i(p) = g - 1$$

Applying the Riemann-Roch there to $D = p + q$, we get:

$$l(D) = 2 - g + (g-1) + 1 = 2$$

Thus, there exist a nonconstant meromorphic function $f \in \mathcal{L}(p+q)$. Thus there is a holomorphic $f : C \to \mathbb{P}^1$where $f^{-1}(\infty) = p + q$. Note that $f^{-1}(\infty)$ cannot e a single point p or q otherwise we get $C \cong \mathbb{P}^1$, contradicting the fact that the genus g of C is ≥ 2. Thus, it must be that:

$$\deg f = 2$$

as we sought to show.

This gives a rough division of Riemann surface of genus $g > 1$:

Definition 4.6: Hyperelliptic and Nonhyperelliptic

Let C be a compact Riemann surface of genus $g \geq 2$. Then it is *hyperelliptic*if there exists a holomorphic mapping of degree 2 $f : C \to \mathbb{P}^1$. Otherwise, C is said to be *nonhyperelliptic*.

We can deal with the nonhyperelliptic example holistically and then see the hyperelliptic examples by examining different genera:

Definition 4.7: Canonical Curve

Let C be a nonhyperelliptic compact Riemann surface of genus $g \geq 2$ with canonical map $\varphi_K : C \to \mathbb{P}^{g-1}$. Then

$$\varphi_K(C) \subseteq \mathbb{P}^{g-1}$$

is called the *canonical curve*.

Note that we put $g \geq 2$, but we shall show shortly that all $g = 2$ compact Riemann surface are hyperelliptic, but this shall not necessarily be the case for $g \geq 3$.

Proposition 4.8: Canonical Curve Smooth

Let C be a nonhyperelliptic compact Riemann surface. Then the canonical map φ_K is injective and $\varphi_K(C)$ is smooth.

***Proof* :**
Proposition 4.5 show's φ_K must be injective, it remains to show φ_K is smooth.

[5, p.136]

4.4 Hyperelliptic Compact Riemann Surfaces

Proposition 4.9: Algebraic Characterization of Hyperelliptic Riemann Surface

Let C be a hyperelliptic compact Riemann surface. Then it can be represented as the normalization of a plane algebraic curve of degree $2g + 2$ where g is the genus of C. By a suitable linear projection, this plane curve will be a 2-sheered covering of a straight line (i.e. $\mathbb{C}$).

***Proof* :**
Let C be a hyperelliptic compact Riemann surface of genus g. We shall construct a holomorphic mapping $f : C \to \mathbb{P}^2$ such that $f(C)$ can be given by a plane algebraic curve of degree $2g + 2$.

As C is hyperelliptic, the exist a degree 2 holomorphic mapping $x : C \to \mathbb{P}^1$. Let R be the ramification divisor of x. By the Riemann-Hurwitz formula (theorem 2.6):

$$\deg R = 2(2 + g - 1) = 2g + 2$$

As x is degree 2, the coefficients of every point in R must be equal to 1, hence we must have:

$$R = p_1 + p_2 + \cdots + p_{2g+2}$$

where each p_i is distinct on C. Now suppose:

$$x^{-1}(\infty) = p + q \qquad x(p_i) = a_i$$

so that x can be visualized as:

FIG HERE

Then as in proposition 4.2 we can define $j : C \to C$ $j(q_1) = q_2$. Just like before, $j \in \mathrm{Aut}(C)$ and $j^2 = 1$. Define again $j^* : K(C) \to K(C)$ by $j^*(a) = a \circ j$, which like before can be extended to $K^1(C)$ and restricted down to $\Omega^1(C)$. Then as $\Omega^1(C) \cong \mathbb{C}^g$, this is a linear involution of C^g with eigenvalues 1 or -1. If the eigenvalue is 1, then the eigenvector corresponding to 1 will be a holomorphic differential form, inducing a nontrivial differential form on $\mathbb{P}^1$ a contradiction. Thus it must be that :

$$j^* : \Omega^1(C) \to \Omega^1(C) \qquad \omega \mapsto -\omega$$

With the setup complete, let us show that such a function does exist by the Riemann-Roch Theorem. Let

$$D = (g+1)p + (g+1)q \in \mathrm{Div}(C)$$

Recall that for nay $\omega \in K^1(C)$ by the Poincaré-Hopf formula $\deg(\omega) = 2g-2 < \deg G$, and so it cannot be that $(\omega) - D \geq 0$, and so $i(D) = 0$. Then by Riemann-Roch:

$$l(D) = (2g+2) - g + 1 = g + 3$$

Thus

$$\mathcal{L}(D) \cong \mathbb{C}^{g+3}$$

As $x(p) = x(q) = \infty$, it must be that $j(D) = D$, so j^* can be seen as a linear transformation on $\mathcal{L}(D)$. Then a $(j^*)^2 = 1$ and j^* also has eigenvalues ± 1, we can split $\mathcal{L}(D)$ into eigenspaces:

$$\mathcal{L}(D) = \mathcal{L}(D)^+ \oplus \mathcal{L}(D)^-$$

where $\mathcal{L}(D)^+$ and $\mathcal{L}(D)^-$ are the eigenspace for eigenvalue $+1$ and -1 respectively. Note that the meromorphic functions in $\mathcal{L}(D)^+$ can be factored as the composition of x and a meromorphic function on $\mathbb{P}^1$ with pole poles only at p, q, and the order of the pole at either point not exceeding $g+1$. Thus, as meromorphic functions on C, the functions $1, x, x^2, ..., x^{g+1}$ form a basis for $\mathcal{L}(D)^+$. Thus:

$$\dim \mathbb{C}(D)^+ = g + 2$$

As $l(D) = g+3$, it follows that

$$\dim \mathbb{C}(D)^- = 1$$

In particular, there exists a meromorphic function $y \in \mathcal{L}(D)$ such that

$$j^*(y) = -y$$

With this build-up, let us compare divisors to prove that there exists a function $g(x) = (x - a_1)(x-a_2)\cdots(x-a_{2g+2})$ such that

$$y^2 = cg(x)$$

where $0 \neq c \in \mathbb{C}$. Let's first find (y^2), in particular it suffices to find (y). since

$$j^*(y) = -y \qquad j(p_i) = p_i \qquad (i = 1, 2, ..., 2g+2)$$

we get:

$$y(p_i) = -y(p_i)$$

and so

$$y(p_i) = 0$$

Thus

$$\deg(y)_0 \geq \deg\left(\sum_{i=1}^{2g+2} p_i\right) = 2g+2$$

But by the Residue Theorem

$$\deg(y)_0 = \deg(y)\infty = 2g+2$$

Thus
$$(y)_0 = \sum_{i=1}^{2g+2} p_i \qquad (y)_\infty = D$$
For $(g(x))$, by definition $(x)_\infty = p + q$ and $(x - a_i)_\infty = p + q$ for all $i = 1, 2..., 2g + 2$. Thus:
$$(g(x))_\infty = (2g + 2)p + (2g + 2)q = 2D$$
Giving the pole divisor. For the root divisor, suppose z_i are local coordinate near p_i for each i where $z_i(p_i) = 0$. Then:
$$x - a_i = z_i^2 h_i(z_i)$$
where $h_i(0) \neq 0$ is holomorphic. But then:
$$(g(x))_0 = \sum_{i=1}^{2g+2} 2p_i$$
and so $(y^2) = (g(x))$, giving that $y^2/g(x)$ is holomorphic on C, which as is compact makes $y^2/g(x) = c \neq 0$ constant. By replacing y with $y/\sqrt{c}$ we get
$$y^2 = g(x)$$
From this, we can define the map:
$$\begin{aligned} f : C &\to \mathbb{P}^2 \\ t &\mapsto [1 : x(t) : y(t)] \\ p, q &\mapsto [0 : 0 : 1] \end{aligned}$$
As C is connected, the solution to an algebraic equation is irreducible (hence connected), f is holomorphic (hence an open map) and C is compact (hence f is a closed map), it must be that f is surjective.

Finally, it is easy to see that each a_I are distinct, hence C is smooth in $\mathbb{C}^2 = \{[1 : x : y] \in \mathbb{P}^2\}$ but singular at $[0 : 0 : 1]$ (which distinguishes from the case of $g = 1$). A linear projection $(x, y) \to x$ gives the 2-sheered covering, giving everything asked for by the proposition, completing the proof.

Proposition 4.10: Canonical Map – Hyperelliptic Case

Let C be a hyperelliptic curve and let x, y be as in proposition 4.9. Then
$$\omega_1 = \frac{dx}{y},\ \omega_2 = \frac{(xdx)}{y},\ ...,\ \omega_g = \frac{(x^{g-1}dx)}{y}$$
forms a basis for $\Omega^1(C)$.

***Proof* :**
[5, p.142]

We thus get a very simple expression for the canonical map:

$$\begin{aligned}\varphi_K : C &\to \mathbb{P}^{g+1} \\ t &\mapsto [1 : x(t) : \cdots : x(t)^{g-1}]\end{aligned}$$

Hence, φ_K factors in the following way:

$$\begin{tikzcd} & \mathbb{P}^{g-1} \\ C \arrow[ur, "\varphi_k"] \arrow[r, "x"'] & \mathbb{P}^1 \arrow[u, "r"'] \end{tikzcd}$$

where

$$\begin{aligned}r : \mathbb{P}^1 &\to \mathbb{P}^{g-1} \\ x &\mapsto [1 : x : \cdots : x^{g-1}]\end{aligned}$$

Note that the converse of proposition 4.9 is true: given any $2g+2$ mutually distinct complex numbers $a_1, a_2, ..., a_{2g+2}$, and letting:

$$C' = \{y^2 = \prod i = 1^{2g+2}(x - a_i) = 0\} \cup [0 : 0 : 1] \subseteq \mathbb{P}^2$$

then the normalization of C of C' is a hyperelliptic compact Riemann surface of genus g, giving an equivalence. Combining this with the commutative diagram given above, we see that C is uniquely determined by the equivalence of vectors in $\mathbb{C}^{2g+2}$ with (pairwise) distinct coordinates:

$$\{(a_1, ..., a_{2g+2})\}$$

where $(a_1, ..., a_{2g+2}) \sim (b_1, ..., b_{2g+2})$ if each coordinate is related by the same linear automorphisms of $\mathbb{P}^1$, that is there exists $a, b, c, d \in \mathbb{C}$ $(ad - bc \neq 0)$ such that

$$\frac{aa_i + b}{ca_i + d} = b_i \qquad (i = 1, 2, ..., 2g + 2)$$

(recall the classification of $\mathrm{Aut}(\mathbb{P}^1)$). Naturally, if replaced by $\lambda a, \lambda b, \lambda c, \lambda d$ the result is unchanged, hence without loss of generality we may assume one of them is fixed.

4.5 Genus 2 always hyperelliptic

Proposition 4.11: Genus 2 is Hyperelliptic

Let C be a compact Riemann surface of genus 2. Then C is hyperelliptic.

***Proof* :**
Let C be a compact Riemann surface of genus 2. Then by proposition 4.5 it suffices to show that φ_K is not injective. But if $\varphi_K : C \to \mathbb{P}^{g-1} = \mathbb{P}^1$ were injective, C would have genus 0, completing the proof.

Thus, we always get C in $\mathbb{P}^1$ as:

$$y^2 = \prod_{i=1}^{6}(x - a_i)$$

for pairwise distinct $a_1, .., a_6$ and forms

$$\omega_1 = \frac{dx}{y} \qquad \omega_2 = \frac{xdx}{y}$$

where:

$$\varphi_: C \to \mathbb{P}^1 \qquad t \mapsto [\omega_1 : \omega_2] = [1 : x(t)]$$

4.6 Genus 3

We shall see in (it says at [5, p.212]) that almost all genus 3 curves are canonical curves (it will be because any smooth forth-degree plane algebraic curve (where $2g - 2 = 4$) has genus $g = 3$ and hence is canonical). Let us start by showing that the canonical curve of genus 3 is a smooth plane algebraic curve of degree 3.

let $H \subseteq \mathbb{P}^n$ be a hyperplane given by:

$$L(\xi) = \sum_{i=0}^{n} l_i \xi^i = 0$$

Let C be a smooth algebraic curve in $\mathbb{P}^n$. Let H not contain any irreducible branches of C. Then pick the elements of $\mathrm{Div}(C)$ in the following way. Choose a generic linear function:

$$L'(\xi) = \sum_{i=0}^{n} l'_i \xi^i$$

where by generic we mean:

$$\{L' = 0\} \cap C \cap \{L = 0\} = \emptyset$$

Then we get:

$$\frac{L}{L'} \in K(\mathbb{P}^n) \qquad \frac{L}{L'}|_C \in K(C)$$

Then the set of all the zeros of $(L/L')|_C$ is given a name

We give a name to this set:

Definition 4.12: Hyperplane Section of Riemann Surface

Let C, H be as above. Then:

$$H \cdot C = \left(\frac{L}{L'}\bigg|_C \right)_0$$

is the *hyperplane section* of C.

Intuitively:

$$H \cdot C = C \cap \{L(\xi) = 0\}$$

Certainly this definition is independent of choice of L'.

Definition 4.13: Degree of Smooth Curve

Let C be a smooth curve in $\mathbb{P}^n$. Then $\deg(H \cdot C)$ is called the *degree* of C and is denoted $\deg C$.

This too is well-defined as it is independent of choice of hyperplane H since if there exists another H_1:

$$\deg(H_1 \cdot C) = \deg\left(\frac{L_1}{L'}\bigg|_C\right)_0 = \deg\left(\frac{L_1}{L'}\bigg|_C\right)_\infty = \deg\left(\frac{L}{L'}\bigg|_C\right)_\infty = \deg\left(\frac{L}{L'}\bigg|_C\right)_0 = \deg(H \cdot C)$$

Note that when $n = 2$, then H is a straight line in $\mathbb{P}^2$ not containing any irreducible curve components of C, and so by Bézout's theorem we know the curve degree is just the same as the plane curve degree given earlier.

In this section, we shall identify any nonhyperelliptic compact Riemann surface of genus $g > 1$ with its image under the canonical map, that is the canonical curve in $\mathbb{P}^{g-1}$ which is a smooth surface. We can do this as C is nonhyperelliptic and hence the canonical map is injective.

Proposition 4.14: Degree and Genus

Let C be a canonical curve of genus g. Then:

$$\deg C = 2g - 2$$

***Proof* :**
Let $\omega_1, ..., \omega_g$ be a basis for $\Omega^1(C)$ so that

$$C = \{[1w_1(p), ..., \omega_g(p)] \mid p \in C\}$$

Suppose:

$$H = \left\{\sum_{i=0}^{g-1} l_i \xi^i = 0\right\} \qquad H' = \left\{\sum_{i=0}^{g-1} l'_i \xi^i = 0\right\}$$

Then by going through the definition we see that $H \cdot C = (\omega)_0$ where $\omega = \sum_{i=0}^{g-1} l_i \omega_i$. As $\omega \in \Omega^1(C)$, $(\omega)_0 = (\omega)$, and hence:

$$H \cdot C = (\omega)$$

Thus

$$\deg C = \deg(H \cdot C) = \deg(\omega) = 2g - g$$

as we sought to show.

Proposition 4.15: Genus 3 By Degree 4 Curve

Let C be a canonical curve of genus 3. The n it is a smooth plane algebraic curve of degree 4.

***Proof* :**
It is smooth by proposition 4.5. As the canonical curve of genus 3 must be in $\mathbb{P}^2$, by the proceeding we have:

$$\deg(C) = 2g - 2 = 2 \cdot 3 - 2 = 4$$

as we sought to show.

Note that if F is a degree 4 algebraic curve, then by the genus formula:

$$g = \frac{(d-1)(d-2)}{2} = \frac{3 \cdot 2}{2} = 3$$

From this we can do the same classification trick as in section 4.4 to show that the converse is mostly true (having to take into account that some canonical embeddings are not injective). Thus, it is as to verify that

$$F(x, y, z) = x^4 + y^4 + z^4 + \lambda x^2 y^2 + \mu y^2 z^2 + \nu z^2 x^2$$

is a canonical curve.

4.7 Genus 4

Canonical genus 4 curves have the amazing property that they can be characterized by the intersection of quadric[1] and a cubic hypersurface in $\mathbb{P}^3$.

Just like before, suppose C Is a smooth algebraic curve in $\mathbb{P}^n$ and $H = \{\xi^0 = 0\}$ is a hypersurface. Let $D = H \cdot C \in \mathrm{Div}(C)$. Notice that $\bigoplus_{k=0}^{\infty} \mathcal{L}(kD)$ and $\mathbb{C}[\xi^0, ..., \xi^n]$ are graded rings. Let $F(\xi) \in \mathbb{C}[\xi^0, ..., \xi^n]$ be a homogeneous polynomial and define:

$$r(F(\xi)) = \left.\frac{F(\xi)}{(\xi^0)^k}\right|_C \in \mathcal{L}(k(D))$$

Then extended r linearly to all of $\mathbb{C}[\xi^0, ..., \xi^n]$, we get ring homomorphism:

$$r : \mathbb{C}[\xi^0, ..., \xi^n] \twoheadrightarrow \bigoplus_{k=0}^{\infty} \mathcal{L}(kD)$$

I claimed this map is surjective; this was shown in [2] (but a judicious application of Noether Theorem also works).

I'l finish typing this up later: [5, p.148]

4.8 Abel's Theorem and its Applications

Let C be ea compact Riemann surface of genus $g \geq 1$. Then $H_1(C, \mathbb{Z})$ is a free abelian group of rank $2g$. Choose a canonical basis $\gamma_1, \gamma_2, ..., \gamma_{2g-1}, \gamma_{2g}$ for $H_1(C, \mathbb{Z})$ satisfying the intersection requirements:

$$\gamma_i \cdot \gamma_{g+1} = 1 \qquad \gamma_{g+i} \cdot \gamma_i = -1$$

[1] I've not seen the word "quadric" before. It means a degree 2 hypersurface embedded in D dimensions. Quadratic seems perfectly fine, but it seems that quadric is just a little more specific

for all i, and 0 otherwise; that is it is the usual vertical and horizontal circles around the handles when visualized in $\mathbb{R}^3$. Recall the period vector:

$$\pi_j = \begin{pmatrix} \int_{\gamma_j} \omega_1 \\ \vdots \\ \int_{\gamma_j} \omega_g \end{pmatrix} \in \mathbb{C}^g$$

for $j = 1, ..., 2g$. Recall that we proved that this vector is well-defined as the paths are closed and that the period matrix is nonsingular (proposition 3.14). Thus, with $\mathbb{Z}$-coordinates, we get a lattice in $\mathbb{C}^g$:

$$\Lambda = \left\{ \sum_{j=1}^{2g} m_j \pi_j \;\middle|\; m_j \in \mathbb{Z} \right\} \subseteq \mathbb{C}^g$$

We thus can define the following:

Definition 4.16: Jacobian Variety

Let C be a compact Riemann surface. Then the g-dimensional complex torus $\mathbb{C}^g/\Lambda$ is called the *Jacobian variety* of C and will be denoted $J(C)$.

This variety will be useful for find ending when are divisor principal, give a measure of the lack thereof. Namely, given a $D \in \mathrm{Div}(C)$, does there exist a $f \in K^*(C)$ such that

$$(f) = D$$

First off, as we are working over compact Riemann surfaces, we can look at divisor of degree 0. Let:

$$\begin{aligned} \deg : \mathrm{Div}(C) &\to \mathbb{Z} \\ D &\mapsto \deg D \end{aligned}$$

Let $\mathrm{Div}^0(C) = \ker(\deg)$. Then if $(f) = D$, it must be that $D \in \mathrm{Div}^0(C)$. The converse is not always true, and Abel's theorem will outline exactly how to associate these two.

To that end, let us start by defining a map from $\mathrm{Div}(C)$ to $J(C)$. Let $q \in C$ be any fixed point and let $\omega \in \Omega^1(C)$ be a holomorphic differential. Then for a point $p \in C$, we can define:

$$\int_q^p \omega \tag{2}$$

Sine there is nontrivial winding, the difference of two paths from q to p is a homology class that can be written as

$$\sum_{j=1}^{2g} m_j \gamma_j$$

hence two values of equation (2) will differ by:

$$\sum_{j=1}^{2g} m_j \int_{\gamma_j} \omega$$

By letting ω run through the basis of $\Omega^1(C)$, we get the following:

Definition 4.17: Abel-Jacobi Map

Let C be a compact Riemann surface, $q \in C$. Then the *Abel-Jacobi map*

$$u : \mathrm{Div}(C) \to J(C)$$

is given by:

$$u(D) = \begin{pmatrix} \sum_{i=1}^k n_i \int_q^{p_i} \omega_1 \\ \vdots \\ \sum_{i=1}^k n_i \int_q^{p_i} \omega_g \end{pmatrix}$$

where $D = \sum_{i=1}^k n_i p_i$.

Certainly this is a homomorphism for if $D = \sum_{i=1}^k p_i - \sum_{i=1}^k q_i \in\in \mathrm{Div}^0(C)$ then by the rules of integration bounds we get the homomorphism rule. We shall see that $(f) = D$ for $D \in \mathrm{Div}^0(C)$ if and only if

$$u(D) = 0$$

Or, slightly more generally:

Theorem 4.18: Abel-Jacobi Theorem

Let C be a compact Riemann surface. Then the sequence:

$$K^*(C) \xrightarrow{(-)} \mathrm{Div}^0(C) \xrightarrow{u} J(C) \to 0$$

is exact

We shall not give full proof of this, but [5] proves it. From this we get:

Definition 4.19: Picard Variety

Let C be a compact Riemann surface and $(-) : K^*(C) \to \mathrm{Div}^0(C)$. Then:

$$\mathrm{Pic}\,(C) = \frac{\mathrm{Div}^0(C)}{\mathrm{im}\,((-))}$$

The Abel-Jacobi map then becomes an isomerism.

(a lot is done here in proving it, rather interesting!!)

(the Riemann bilinear relations puts important constrictions on the period matrix that shows how it reflects some of its topological properties!) (this gives us:)

Theorem 4.20: Torelli Theorem

Let C, C' be compact Riemann surfaces. Then a necessary and sufficient conditions for $C \cong C'$ as biholomorphisms is that the have the same normalized period matrix under a suitable choice of canonical homology bases

***Proof* :**
[5, p.174]

(then it goes onto prove Jacobi inversion theorem; that $u : \mathrm{Div}^0(C) \to J(C)$ is surjective)

4.9 Applications of Abel's Theorem

Let C be a nonhyperelliptic compact Riemann surface of genus 3; without loss of generality we may identify C with its canonical curve in $\mathbb{P}^2$. The nC is a fourth-degree smooth plane algebraic curve. Let L be a straight line in $\mathbb{P}^2$. Then:

$$D = L \cdot C = p_1 + p_2 + p_3 + p_4 \in C^{(4)}$$

is a hyperplane section. Let $E \in \mathbb{C}^{(4)}$. Then $u(E) = u(D)$ if and only if there exists a line L' such that

$$E = L' \cdot C$$

hence, E must be a hyperplane. Indeed, if $E = L' \cdot C$, let L and L' be given by linear equations l, l' so that $L = \{l = 0\}$ and $L' = \{l' = 0\}$. Letting:

$$f = \left.\frac{l'}{l}\right|_C \in K(C)$$

then $(f) = E - D$, and so $0 = u(E) - u(D)$, hence $u(E) = u(D)$. Conversely, if $u(D) = u(E)$, then we need to show that:

$$E \equiv q_1 + q_2 + q_3 + q_4$$

is a hyperplane section. Suppose that L' is the unique line passing through q_1, q_2. Then:

$$L' \cdot C = q_1 + q_2 + q' + q''$$

If $L' \cdot C \neq E$, then it must be that :

$$q_3 + q_4 - q' - q'' \neq 0$$

If $u(D) = u(E)$, then $u(E) = u(L' \cdot C)$ (as D and $L' \cdot C$ are both hyperplanes sections), and so:

$$u(q_3 + q_4 - q' - q'') = 0$$

Thus by the Abel-Jacobi Theorem there exists an f such that

$$(f) = q_3 + q_4 - q' - q''$$

Now, consider $f : C \to \mathbb{P}^1$ as a holomorphic mapping. Then f has either degree 1 or 2 (depending on water one of $\{q_3, q_3\}$ is equal to one of $\{q', q''\}$). Implying that C is either Riemann sphere or a hyperelliptic Riemann surface, but that contradicts the fact that C is nonhyperelliptic of genus 3. Thus:

$$E = L' \cdot C$$

Thus, given a hyperplane section D of a canonical curve of genus 3, $|D|$ can be identified with the set of hyperplanes of $\mathbb{P}^2$, that is:

$$|D| \cong \mathbb{P}^2$$

Giving us a full geometric picture!

Next, if $g = 1$, we can rather easily classify all such surfaces up to biholomorphism. We first have:

Theorem 4.21: Jacobi Variety in Genus 1

Let C be a compact Riemann surface of genus 1. Then it is biholomorphic to its Jacobian variety

$$C \cong J(C)$$

Given a normalized period matrix

$$\Pi = (1\ z)1 \times x$$

Then $C \cong \mathbb{C}/\Lambda$, where Λ is the lattice spanned by 1 and z.

Proof **:**
In [5, p.193], this was a consequence in the discussion of the Jacobi inversion theorem in the case of $d = 1$, where immediately concludes:

$$u : C \to J(C)$$

is a biholomorphic mapping.

Theorem 4.22: Up To Projective Special Automorphism

Let $C = \mathbb{C}/\Lambda$ and $C' = \mathbb{C}/\Lambda'$ where:

$$\begin{aligned} \Lambda &= \{m + nz \mid m, n \in \mathbb{Z}\} && \mathrm{im} z > 0 \\ \Lambda' &= \{m + nz' \mid m, n \in \mathbb{Z}\} && \mathrm{im} z' > 0 \end{aligned}$$

hen $C \cong C'$ if and only if and only if

$$z = \frac{\alpha z' + \beta}{\gamma z' + \delta}$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and

$$\alpha\delta - \beta\gamma = 1$$

Proof **:**
In [5, p.194], but can also be found in [4] or [1].

(then he goes onto finish off with some commentary about elliptic integrals, the "usual" representation a $p(x) = 4x^3 + ax + b$, uses Bézouts theorem to get the number of flex points of an nth degree curve (that is $3n(n-2)$ counting multiplicity), and he uses all of this to show the original connection to integrating ellipses. The last thing he does is show some algebraic properties of $x(u)$; just like was shown a Weierstrass equation section in a complex analysis course).

5 Riemann Roch in Modern Language

here

6 Generalized Riemann Roch: Vector Bundles

Let X be a complex irreducible algebraic variety (not necessarily smooth). Define:

1. X_{Zar}: X with the Zariski topology (abbreviated to Z-topology) and $\mathcal{O}_X$ the germs of algebraic functions
2. $X_{\mathbb{C}}$: X with the complex topology (abbreviated to $\mathbb{C}$-topology)and $\mathcal{O}_X$ the germs of algebraic functions
3. X^{an}: X with the complex topology and $\mathcal{O}_X$ the germs of holomorphic functions
4. X_{smooth}: X with the complex topology and $\mathcal{O}_X$ the germs of C^∞-functions

Then a vector bundle $p : E \to X$ can either be locally trivial in the $\mathbb{Z}$-topology or the $\mathbb{C}$-topology. As X is irreducible, it is connected, so the rank is a well-defined. If $E \to X$ has rank r, then:

1. (covering) there exists an open covering $\{U_\alpha \to X\}_\alpha$ along with isomorphisms:

$$\varphi_\alpha : p^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^r$$

   We shall sometimes denote $p^{-1}(U_\alpha)$ as $E|_{U_\alpha}$.

2. (local trivialization) For every α, the following diagram commutes:

$$\begin{tikzcd} p^{-1}(U_\alpha) \arrow[rr, "\sim"] \arrow[dr, "p"'] & & U_\alpha \times \mathbb{C}^r \arrow[dl, "\pi_1"] \\ & X & \end{tikzcd}$$

3. (gluing – cocycle condition)

$$\begin{tikzcd}
p^{-1}(U_\alpha) \arrow[r, "\phi_\alpha"] \arrow[d] & U_\alpha \prod \mathbb{C}^r \arrow[d] \\
p^{-1}(U_\alpha \cap U_\beta) \arrow[r, "\phi_\alpha"] \arrow[d, equal] & (U_\alpha \cap U_\beta) \prod \mathbb{C}^r \arrow[d, "g_\alpha^\beta"] \\
p^{-1}(U_\beta \cap U_\alpha) \arrow[r, "\phi_\beta"] \arrow[d] & (U_\beta \cap U_\alpha) \prod \mathbb{C}^r \arrow[d] \\
p^{-1}(U_\beta) \arrow[r, "\phi_\beta"] & U_\beta \prod \mathbb{C}^r
\end{tikzcd}$$

where by commutativity $g_\alpha^\beta = \varphi_\beta \circ \varphi_\alpha^{-1}|_{p^{-1}(U_\alpha \cap U_\beta)}$.

Note then that

$$g_\alpha^\beta|_{U_\alpha \cap U_\beta} = \mathrm{id} \qquad \text{and} \qquad g_\alpha^\beta|_{\mathbb{C}^r} \in \mathrm{GL}_r(\Gamma(U_\alpha \cap U_\beta), \mathcal{O}_X)$$

On a triple overlap, we get:

$$g_\beta^\gamma \circ g_\alpha^\beta = g_\alpha^\beta \qquad \text{and} \qquad (g_\beta^\alpha)^{-1} = g_\alpha^\beta$$

Hence, the collection $\{g_\alpha^\beta\}$ forms a 1-cocycle in $Z^1(\{U_\alpha \to X\}, \mathbb{GL})$ where $\mathbb{GL}$ or $\mathbb{GL}(X)$ is:

$$\Gamma(U, \mathbb{GL}_r(X)) = \mathrm{GL}_r(\Gamma(U, \mathcal{O}_X))$$

where the right hand side is the group of invertible linear maps on the free module $\Gamma(U, \mathcal{O}_X)^r \cong \Gamma(U, \mathcal{O}_X^r)$. When $r = 1$, we often denote $\mathbb{GL}_1(X)$ as $\mathbb{G}_m$ or $\mathcal{O}_X^*$.

Let's talk about choice. Say $\{\psi_\alpha\}_\alpha$ is another trivialization. Without loss of generality we may refine each cover so that $\{\varphi_\alpha\}_\alpha$ and $\{\psi_\alpha\}_\alpha$ use the same cover. Then there exists an isomorphism: $\sigma_\alpha : U_\alpha \times \mathbb{C}^r \to U_\alpha \times \mathbb{C}^r$ where the following diagram commutes:

$$\begin{tikzcd}
 & U_\alpha \times \mathbb{C}^r \arrow[dd, "\sigma_\alpha"] \\
p^{-1}(U_\alpha) \arrow[ur, "\varphi_\alpha"] \arrow[dr, "\psi_\alpha"'] & \\
 & U_\alpha \times \mathbb{C}^r
\end{tikzcd}$$

Then $\{\sigma_\alpha\}_\alpha$ form 0-cochains in $C^0(\{U_\alpha \to X\}, \mathbb{GL}_r)$. if $\{h_\alpha^\beta\}$ is the gluing information for the $\{\psi_\alpha\}$ then:

$$\phi_\beta = g_\alpha^\beta \circ \phi_\alpha \psi_\beta \qquad\qquad = h_\alpha^\beta \circ \psi_\alpha \psi_\alpha = \sigma_\alpha \circ \phi_\alpha.$$

Thus: $\sigma_\beta \circ \phi_\beta = \psi_\beta = h_\alpha^\beta \circ \sigma_\alpha \circ \phi_\alpha$, so:

$$\phi_\beta = (\sigma_\beta^{-1} \circ h_\alpha^\beta \circ \sigma_\alpha) \circ \phi_\alpha,$$

and so the transition between g_α^β and h_α^β is:

$$g_\alpha^\beta = \sigma_\beta^{-1} \circ h_\alpha^\beta \circ \sigma_\alpha$$

We thus get a conjugation equivalence relation $\sim$ on $Z^1(\{U_\alpha \to X\}, \mathbb{GL}_r)$ which trivializes co-cycles:

$$Z^1/\sim = H^1(\{U_\alpha \to X\}, \mathbb{GL}_r)$$

This is not a group as $\mathbb{GL}_r$ is not abelian, however it is a pointed set (the trivial bundle is singled out, pointed out). By passing through the limit over the covers by refinement, we can label the limit $\check{\mathrm{H}}^1(X, \mathbb{GL}_r)$ and we then have:

Theorem 6.1: Vector Bundle Classification Through Cohomology

Let X be an algebraic curve of type (I)-(IV). Then the set of isomorphisms classes of rank r of vector bundles is isomorphic to $\check{\mathrm{H}}^1(X, \mathbb{GL}_r)$

***Proof* :**
the above work

This group is indeed the right choice, for if $1 \to G' \to G \to G'' \to 1$ is an exact sequence of of sheaves of groups (not necessarily commutative), then

$$\begin{tikzcd} 1 \arrow[r] & G'(X) \arrow[r] & G(X) \arrow[r] & G''(X) \arrow[dll, rounded corners, "\delta_0"] \\ & \check{H}^1(X, G') \arrow[r] & \check{H}^1(X, G) \arrow[r] & \check{H}^1(X, G'') \end{tikzcd}$$

is an exact sequence of pointed sets. To compute $\delta_0(\sigma)$ where $\sigma \in G''(X)$, proceed as follows: Cover X by suitable U_α and pick $s_\alpha \in G(U_\alpha)$ mapping to $\sigma|U_\alpha$ in $G''(U_\alpha)$. Set

$$\delta_0(\sigma) = s_\alpha s_\beta^{-1} \text{ on } U_\alpha \cap U_\beta / \sim .$$

Then $\delta_0(\sigma) \in \check{\mathrm{H}}^1(X, G')$.

Note that we took $\mathbb{GL}_r(X)$, but we could have taken the automorphisms of any object, namely if F is an object and $\mathrm{Aut}(F)$ is the automorphisms, we then call the X-torsor for F an "object E over X" locally (on X) of the form $U \times F$ and glued by paired (id, g) where $g \in \mathrm{Mor}(U \cap V, \mathrm{Aut}(F))$. Then we just classified all X-torsors of F.

(3)

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathbb{G}_{r+1} \longrightarrow \mathbb{PGL}_r \longrightarrow 0 \text{ is exact.}$$

I'll finish later. the point is that we can look at iso of vect bundles over either Zariski or Complex topology where $\mathscr{F} = \mathcal{O}_X$ is alg functions and it is isomorphic; or holo functions over Zariski topology injects into over complex topology.

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